

► NEUROLOGY—ANATOMY AND PHYSIOLOGY

Cells of the nervous system

Neurons and nonneuronal (glial) cells.
Neurons—permanent, signal-transmitting cells of the nervous system composed of dendrites (receive input), cell bodies, and axons (send output). Dendrites and cell bodies can be seen on Nissl staining (stains rough endoplasmic reticulum [RER]; not present in axons). Markers: neurofilament protein, synaptophysin.

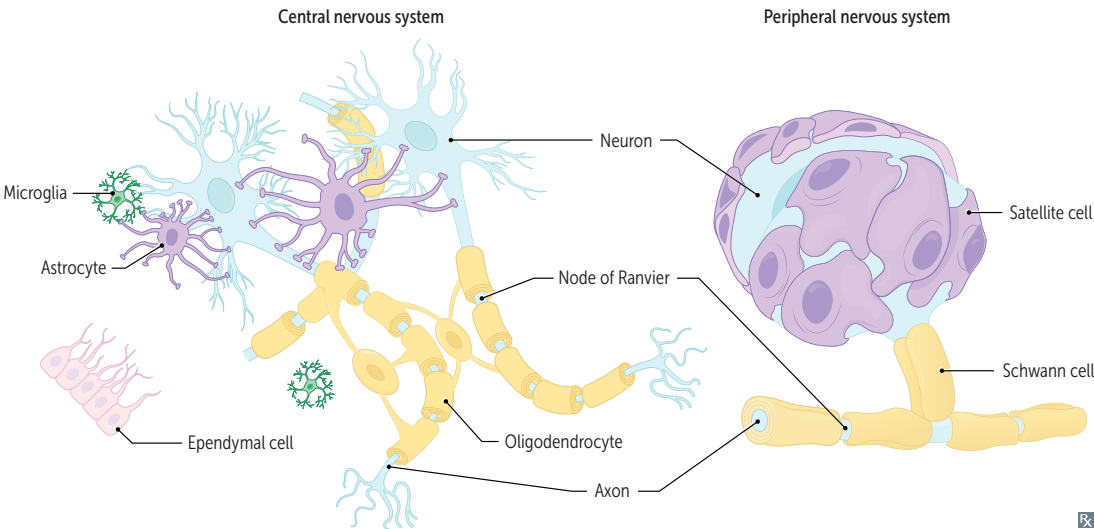
CNS glial cells—neuroectoderm (except microglia, which derive from mesoderm).
PNS glial cells—neural crest ectoderm.
Myelin is a multilayer wrapping of electrical insulation formed around axons
→ ↑ conduction velocity of transmitted signals via saltatory conduction of action potentials at nodes of Ranvier (↑↑ Na⁺ channel density).

CNS glial cells

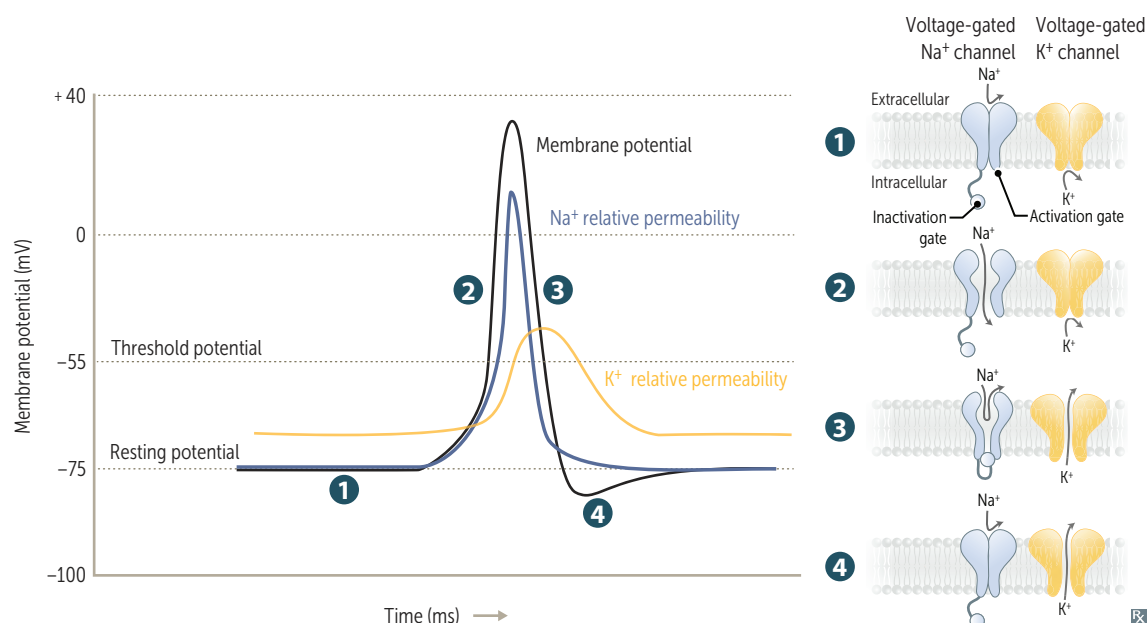
Astrocytes	Physical support, repair, removal of excess neurotransmitters, component of blood-brain barrier, glycogen fuel reserve buffer. GFAP ⊕.	Largest and most abundant glial cell in CNS. Reactive gliosis in response to neural injury.
Oligodendrocytes	Myelinate axons in CNS (including CN II). “Fried egg ” appearance histologically (“ oleggodendrocytes ”).	Each myelinates many axons (~ 30). Predominant type of glial cell in white matter. Injured in multiple sclerosis, leukodystrophies, progressive multifocal leukoencephalopathy.
Ependymal cells	Ciliated simple columnar glial cells lining ventricles and central canal of spinal cord. Apical surfaces are covered with cilia (which circulate CSF) and microvilli (which help with CSF absorption).	Specialized ependymal cells (choroid plexus) produce CSF.
Microglia	Activation in response to tissue damage → release of inflammatory mediators (eg, nitric oxide, glutamate). Not readily discernible by Nissl stain.	Phagocytic scavenger cells of CNS. HIV-infected microglia fuse to form multinucleated giant cells in CNS in HIV-associated dementia.

PNS glial cells

Satellite cells	Surround neuronal cell bodies in ganglia.	Similar supportive role to astrocytes.
Schwann cells	Myelinate axons in PNS (including CN III-XII). S100 ⊕.	Each myelinates a single axon (“ Schwone ”). Injured in Guillain-Barré syndrome.



Neuron action potential

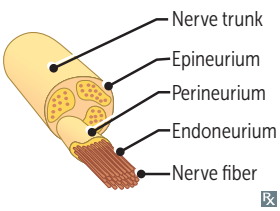


- 1 Resting membrane potential: membrane is more permeable to K^+ than Na^+ at rest. Voltage-gated Na^+ and K^+ channels are closed.
- 2 Membrane depolarization: Na^+ activation gate opens $\rightarrow$ Na^+ flows inward.
- 3 Membrane repolarization: Na^+ inactivation gate closes at peak potential, thus stopping Na^+ inflow. K^+ activation gate opens $\rightarrow$ K^+ flows outward.
- 4 Membrane hyperpolarization: K^+ activation gates are slow to close $\rightarrow$ excess K^+ efflux and brief period of hyperpolarization. Voltage-gated Na^+ channels switch back to resting state. Na^+/K^+ pump restores ions concentration.

Sensory receptors

RECEPTOR TYPE	SENSORY NEURON FIBER TYPE	LOCATION	SENSES
Free nerve endings	$A\delta$ —fast, myelinated fibers C —slow, unmyelinated A Delta plane is fast, but a taxC is slow	All tissues except cartilage and eye lens; numerous in skin	Pain, temperature
Meissner corpuscles	Large, myelinated fibers; adapt quickly	Glabrous (hairless) skin (eg, palms, soles, lips)	Dynamic, fine/light touch, low-frequency vibration, skin indentation
Pacinian corpuscles	Large, myelinated fibers; adapt quickly	Deep skin layers, ligaments, joints	High-frequency vibration, pressure (pressure cooker)
Merkel discs	Large, myelinated fibers; adapt slowly	Finger tips, superficial skin	Pressure, deep static touch (eg, shapes, edges)
Ruffini corpuscles	Large, myelinated fiber intertwined among collagen fiber bundles; adapt slowly	Finger tips, joints	Stretch, joint angle change

Peripheral nerve



Endoneurium—thin, supportive connective tissue that ensheathes and supports individual myelinated nerve fibers. May be affected in Guillain-Barré syndrome.

Perineurium (blood-nerve permeability barrier)—surrounds a fascicle of nerve fibers.

Epineurium—dense connective tissue that surrounds entire nerve (fascicles and blood vessels).

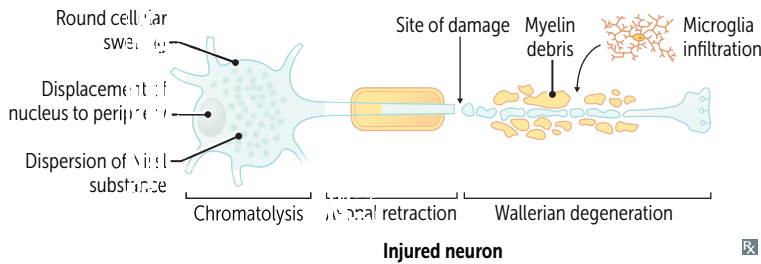
Endo = inner
Peri = around
Epi = outer

Neuronal response to axonal injury

Chromatolysis—dispersion of Nissl substance throughout cytoplasm (RER no longer visible on staining). Neuronal cell body reaction reflecting ↑ protein synthesis in effort to repair damaged axon. Accompanied by round cellular swelling and displacement of nucleus to periphery.

Axonal retraction—proximal axon segment retracts and sprouts new protrusions that grow toward other neurons for potential reinnervation. In PNS, Schwann cells create a tract that guides axonal regeneration.

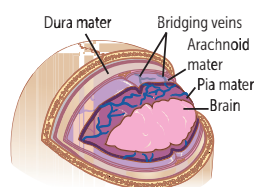
Wallerian degeneration—distal axon segment and associated myelin sheath disintegrates with macrophages removing debris. In CNS, persistence of myelin debris and reactive gliosis prevent axonal regeneration. Facilitate axonal regeneration in response to neural injury.



Neurotransmitter changes with disease

	LOCATION OF SYNTHESIS	ANXIETY	DEPRESSION	SCHIZOPHRENIA	ALZHEIMER DISEASE	HUNTINGTON DISEASE	PARKINSON DISEASE
Acetylcholine	Basal nucleus of Meynert (forebrain)				↓	↓	↑
Dopamine	Ventral tegmentum, SNc (midbrain)		↓	↑		↑	↓
GABA	Nucleus accumbens (basal ganglia)	↓				↓	
Norepinephrine	Locus ceruleus (pons)	↑	↓				
Serotonin	Raphe nuclei (brainstem)	↓	↓				↓

Meninges



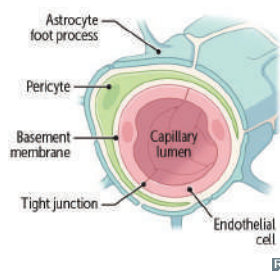
Three membranes that surround and protect the brain and spinal cord. Derived from both neural crest and mesoderm:

- Dura mater—thick outer layer closest to skull.
- Arachnoid mater—middle layer, contains weblike connections.
- Pia mater—thin, fibrous inner layer that firmly adheres to brain and spinal cord.

CSF flows in the subarachnoid space, located between arachnoid and pia mater.

Epidural space—potential space between dura mater and skull/vertebral column containing fat and blood vessels. Site of blood collection associated with middle meningeal artery injury.

Blood-brain barrier



Prevents circulating blood substances (eg, bacteria, drugs) from reaching the CSF/CNS. Formed by 4 structures:

- Tight junctions between nonfenestrated capillary endothelial cells
- Basement membrane
- Pericytes
- Astrocyte foot processes

Glucose and amino acids cross slowly by carrier-mediated transport mechanisms.

Nonpolar/lipid-soluble substances cross rapidly via diffusion.

Circumventricular organs with fenestrated capillaries and no blood-brain barrier allow molecules in blood to affect brain function (eg, area postrema—vomiting after chemotherapy; OVLT [organum vasculosum lamina terminalis]—osmoreceptors) or neurosecretory products to enter circulation (eg, neurohypophysis—ADH release).

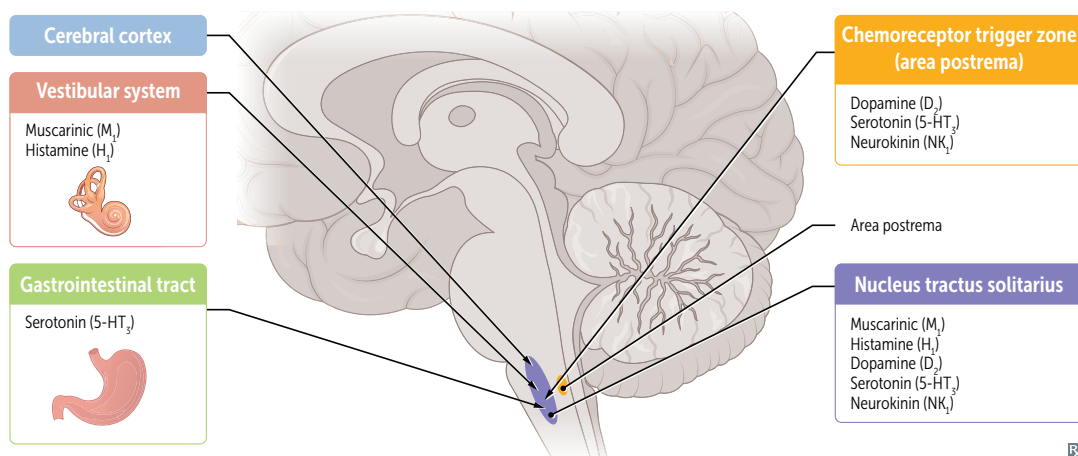
BBB disruption (eg, stroke) → vasogenic edema. Hyperosmolar agents (eg, mannitol) can disrupt the BBB → ↑ permeability of medications.

Vomiting center

Coordinated by NTS in the medulla, which receives information from the chemoreceptor trigger zone (CTZ, located within area postrema (pronounce “puke”-strema) at the base of the 4th ventricle), GI tract (via vagus nerve), vestibular system, and CNS.

CTZ and adjacent vomiting center nuclei receive input through 5 major receptors: histamine (H_1), muscarinic (M_1), neurokinin (NK-1), dopamine (D_2), and serotonin ($5-HT_3$).

- $5-HT_3$, D_2 , and NK-1 antagonists treat chemotherapy-induced vomiting.
- H_1 and M_1 antagonists treat motion sickness; H_1 antagonists treat hyperemesis gravidarum.



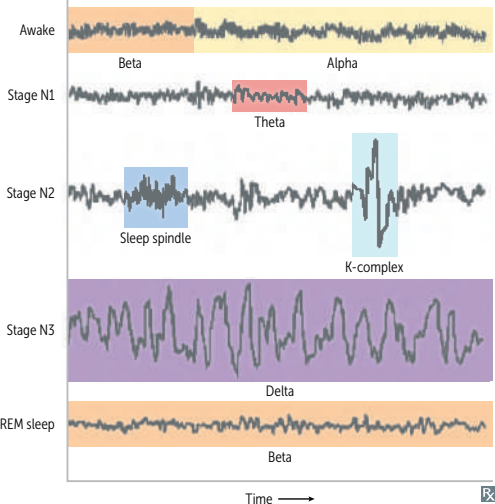
Sleep physiology

Sleep occurs in 4-6 cycles per night, each lasting ~90 mins and consisting of 2 main stages:

- Non-rapid eye movement (NREM) sleep
- Rapid-eye movement (REM) sleep; duration of REM sleep ↑ through the night

Sleep-wake cycle is regulated by circadian rhythm in the suprachiasmatic nucleus (SCN). ↓ light → ↓ SCN activity → ↑ NE from superior cervical ganglion → ↑ melatonin from pineal gland.

EEG waveforms: **B**eta, **A**lpha, **T**heta, **S**leep spindle, **D**elta, **B**eta. At night, **BATS Drink Blood**.

SLEEP STAGE (% OF TOTAL SLEEP)	DESCRIPTION	EEG WAVEFORM
Awake	Alert, active mental concentration. Eyes open—beta waves (highest frequency, lowest amplitude). Eyes closed—alpha waves.	
NREM sleep		
Stage N1 (5%)	Light sleep; theta waves.	
Stage N2 (45%)	Deeper sleep; sleep spindles and K complexes. When bruxism occurs (“ twoth ” grinding in N2).	
Stage N3 (25%)	Deepest NREM sleep (slow-wave sleep); delta waves (lowest frequency, highest amplitude). When bedwetting , sleepwalking , and night terrors occur (wee and flee in N3).	
REM sleep (25%)	Loss of muscle tone (atonia) except in diaphragm and extraocular muscles, ↑ brain O ₂ use, variable pulse/BP. When dreaming , nightmares, and penile/clitoral tumescence occur (REMember dreams).	

REM sleep behavior disorder

Loss of atonia leading to dream enactment (often violent) and vocalization. Most commonly associated with Lewy body dementia and Parkinson disease.

Factors affecting sleep architecture

Alcohol, benzodiazepines, barbiturates: ↓ N3 and REM sleep (benzodiazepines are useful for sleepwalking and night terrors).
Aging: ↓ N3 and REM sleep, ↑ sleep-onset latency, early morning awakening.
Depression: ↓ N3 sleep, ↑ REM sleep, ↓ REM latency, repeated nighttime awakenings, early morning awakening (terminal insomnia).
Narcolepsy: ↓ REM latency.

Hypothalamus

Maintains homeostasis by regulating **T**hirst and water balance, controlling **A**denohypophysis (anterior pituitary) and **N**eurohypophysis (posterior pituitary) release of hormones produced in the hypothalamus, and regulating **H**unger, **A**utonomic nervous system, **T**emperature, and **S**exual urges (**TAN HATS**).

Inputs (areas not protected by blood-brain barrier): OVL_T (senses change in osmolarity), area postrema (found in dorsal medulla, responds to emetics).

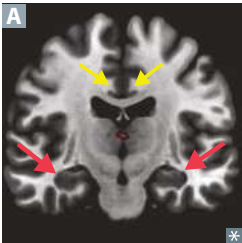
Lateral nucleus	Hunger. Stimulated by ghrelin, inhibited by leptin.	L ateral injury makes you l ean. Destruction → anorexia, failure to thrive (infants).
Ventromedial nucleus	Satiety. Stimulated by leptin.	V entromedial injury makes you v ery m assive. Destruction (eg, craniopharyngioma) → hyperphagia.
Anterior nucleus	Cooling, parasympathetic.	A/C = A nterior C ooling.
Posterior nucleus	Heating, sympathetic.	H eating controlled by p osterior nucleus ("hot pot").
Suprachiasmatic nucleus	Circadian rhythm.	SCN is a S un- C ensing N ucleus.
Supraoptic and paraventricular nuclei	Synthesize ADH and oxytocin.	SAD POX : Supraoptic = ADH , Paraventricular = OXY tocin. ADH and oxytocin are carried by neurophysins down axons to posterior pituitary, where these hormones are stored and released. Destruction → central diabetes insipidus.
Preoptic nucleus	Thermoregulation, sexual behavior. Releases GnRH.	Failure of GnRH-producing neurons to migrate from olfactory pit → Kallmann syndrome.

Thalamus

Major relay for all ascending sensory information except olfaction.

NUCLEI	INPUT	SENSES	DESTINATION	MNEMONIC
Ventral posterolateral nucleus	Spinothalamic and dorsal columns/medial lemniscus	V ibration, p ain, p ressure, p roprioception (conscious), l ight touch, t emperature	1° somatosensory cortex (parietal lobe)	
Ventral postero-medial nucleus	Trigeminal and gustatory pathway	F ace sensation, taste	1° somatosensory cortex (parietal lobe)	V ery p retty m akeup goes on the f ace
Lateral geniculate nucleus	CN II, optic chiasm, optic tract	Vision	1° visual cortex (occipital lobe)	L ateral = l ight (vision)
Medial geniculate nucleus	Superior olive and inferior colliculus of tectum	Hearing	1° auditory cortex (temporal lobe)	M edial = m usic (hearing)
Ventral anterior and ventral lateral nuclei	Basal ganglia, cerebellum	Motor	M otor cortices (frontal lobe)	V enus a stronauts v ow to l ove m oving

Limbic system



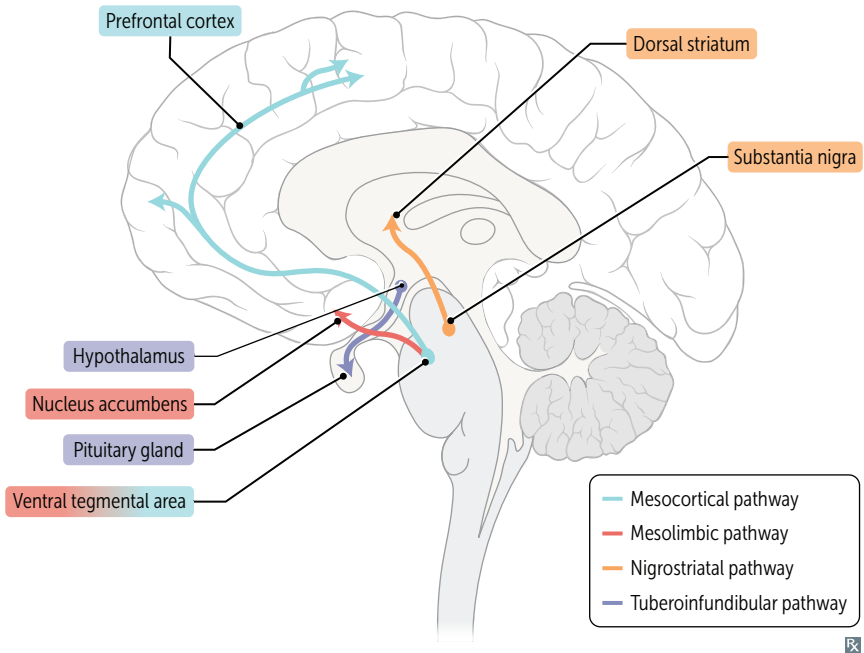
Collection of neural structures involved in emotion, long-term memory, olfaction, behavior modulation, ANS function. Consists of hippocampus (red arrows in **A**), amygdalae, mammillary bodies, anterior thalamic nuclei, cingulate gyrus (yellow arrows in **A**), entorhinal cortex. Responsible for **f**eeding, **f**leeing, **f**ighting, **f**eeling, and **s**ex.

The famous **5 F**'s.

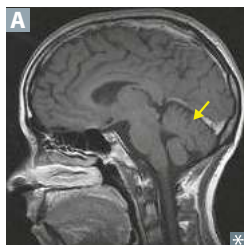
Dopaminergic pathways

Commonly altered by drugs (eg, antipsychotics) and movement disorders (eg, Parkinson disease). The mesocortical and mesolimbic pathways are involved in addiction behaviors.

PATHWAY	PROJECTION	FUNCTION	SYMPTOMS OF ALTERED ACTIVITY	NOTES
Mesocortical	Ventral tegmental area → prefrontal cortex	Motivation and reward	↓ activity → negative symptoms	Antipsychotics have limited effect
Mesolimbic	Ventral tegmental area → nucleus accumbens		↑ activity → positive symptoms	1° therapeutic target of antipsychotics
Nigrostriatal	Substantia nigra → dorsal striatum	Motor control (pronounce “nigro stri deatal”)	↓ activity → extrapyramidal symptoms	Significantly affected by antipsychotics and in Parkinson disease
Tuberoinfundibular	Hypothalamus → pituitary	Regulation of prolactin secretion	↓ activity → ↑ prolactin	Significantly affected by antipsychotics



Cerebellum



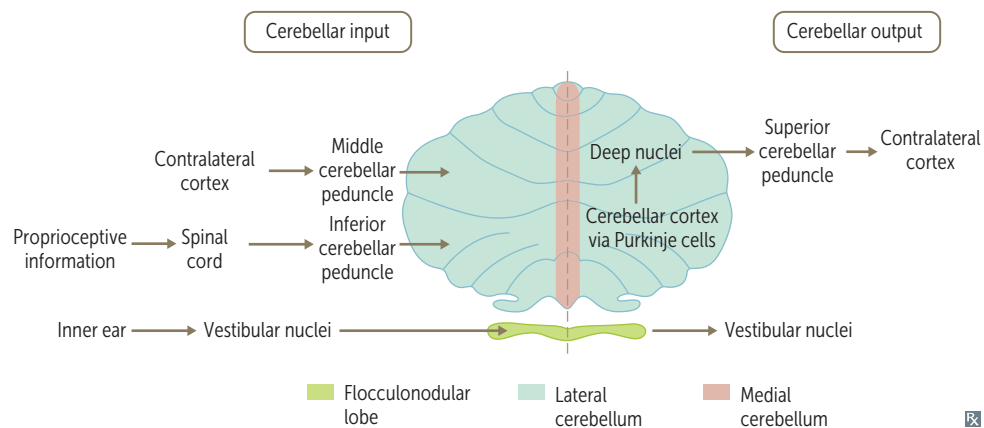
Modulates movement; aids in coordination and balance **A**.

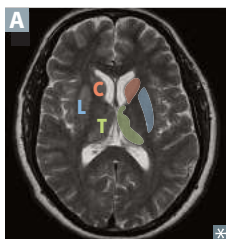
- Ipsilateral (unconscious) proprioceptive information via inferior cerebellar peduncle from spinal cord
- Deep nuclei (lateral → medial)—dentate, emboliform, globose, fastigial (don't eat greasy foods)

Medial cerebellum (eg, vermis) controls axial and proximal limb musculature bilaterally (**medial** structures).

Lateral cerebellum (ie, hemisphere) controls distal limb musculature ipsilaterally (**lateral** structures).

Tests: rapid alternating movements (pronation/supination), finger-to-nose, heel-to-shin, gait, look for intention tremor.



Basal ganglia

Important in voluntary movements and adjusting posture **A**.

Receives cortical input, provides negative feedback to cortex to modulate movement.

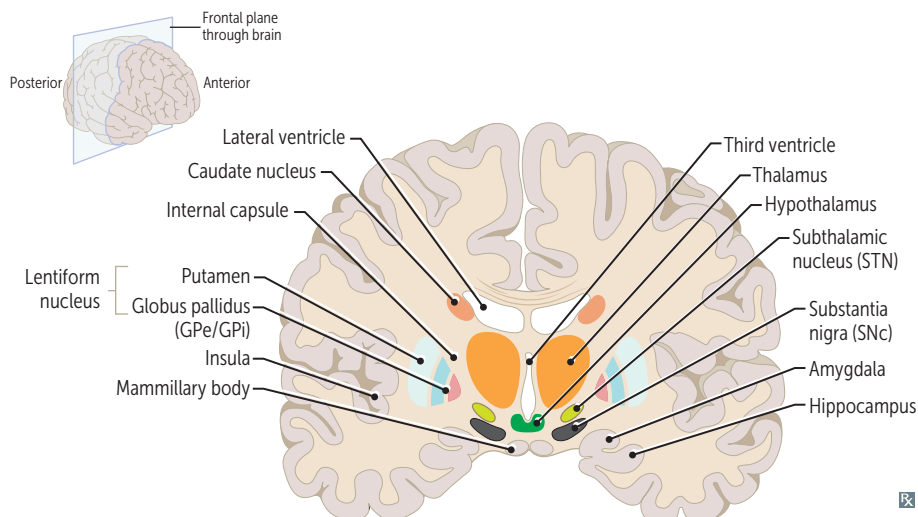
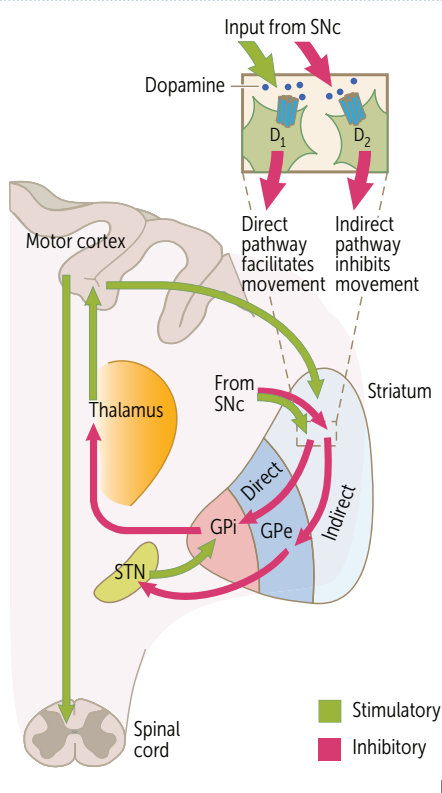
Striatum = putamen (motor) + **C**audate nucleus (cognitive).

Lentiform nucleus = putamen + globus pallidus.

Direct (excitatory) pathway—cortical input (via glutamate) stimulates GABA release from the striatum, which inhibits GABA release from GPi, disinhibiting (activating) the **T**halamus → ↑ motion.

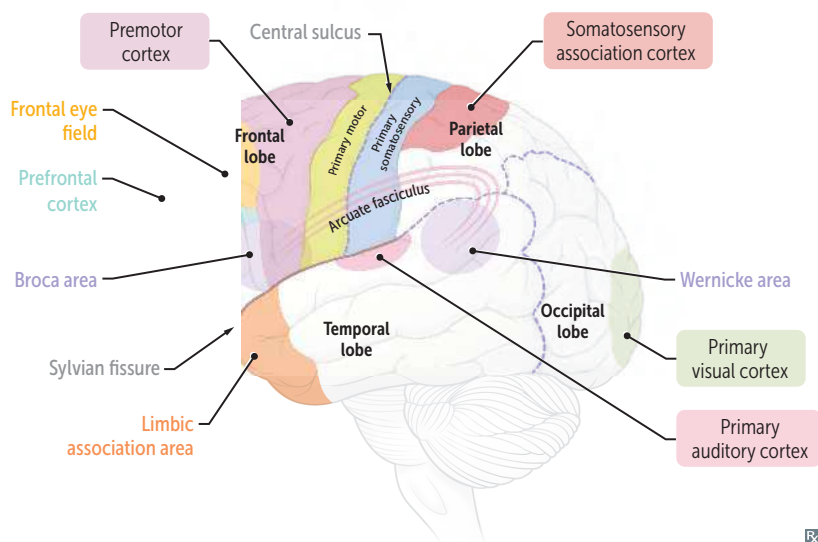
Indirect (inhibitory) pathway—cortical input (via glutamate) stimulates GABA release from the striatum, which inhibits GABA release from GPe, disinhibiting (activating) the STN. STN input (via glutamate) stimulates GABA release from GPi, inhibiting the **T**halamus → ↓ motion.

Dopamine from SNc (nigrostriatal pathway) stimulates the direct pathway (by binding to D_1 receptor) and inhibits the indirect pathway (by binding to D_2 receptor) → ↑ motion.



P2

Cerebral cortex regions



Cerebral perfusion

Relies on tight autoregulation. Primarily driven by P_{CO_2} (P_{O_2} also modulates perfusion in severe hypoxia, eg, mountain sickness—hypoxia $\rightarrow$ $\uparrow$ cerebral vasodilation $\rightarrow$ $\uparrow$ cerebral blood flow $\rightarrow$ cerebral edema).

Also relies on a pressure gradient between mean arterial pressure (MAP) and intracranial pressure (ICP). $\downarrow$ blood pressure or $\uparrow$ ICP $\rightarrow$ $\downarrow$ cerebral perfusion pressure (CPP).

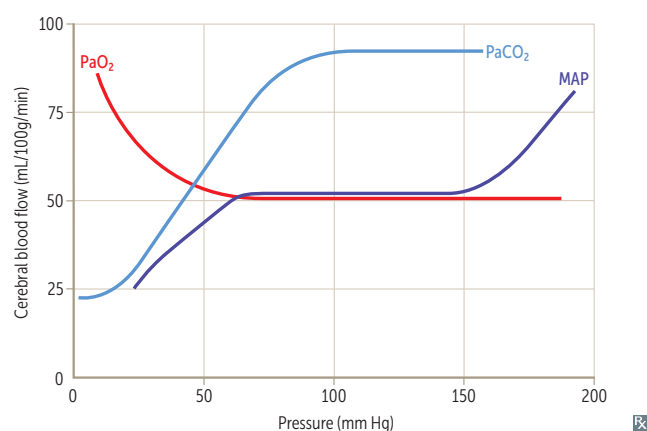
Cushing reflex—triad of hypertension, bradycardia, and respiratory depression in response to $\uparrow$ ICP.

Therapeutic hyperventilation $\rightarrow$ $\downarrow$ P_{CO_2} $\rightarrow$ vasoconstriction $\rightarrow$ $\downarrow$ cerebral blood flow $\rightarrow$ $\downarrow$ ICP. May be used to treat acute cerebral edema (eg, 2° to stroke) unresponsive to other interventions.

CPP = MAP – ICP. If CPP = 0, there is no cerebral perfusion $\rightarrow$ brain death (coma, absent brainstem reflexes, apnea).

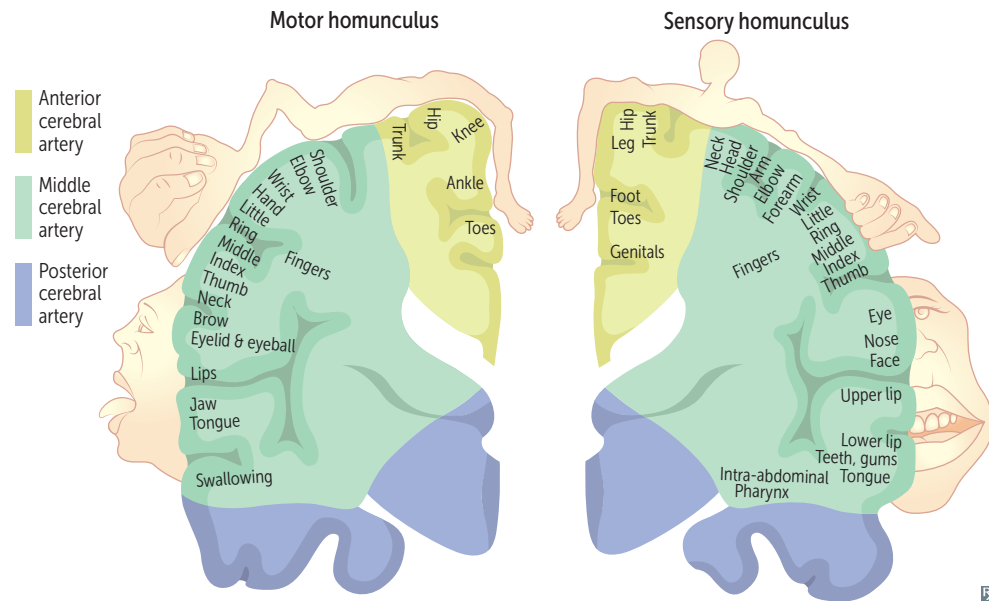
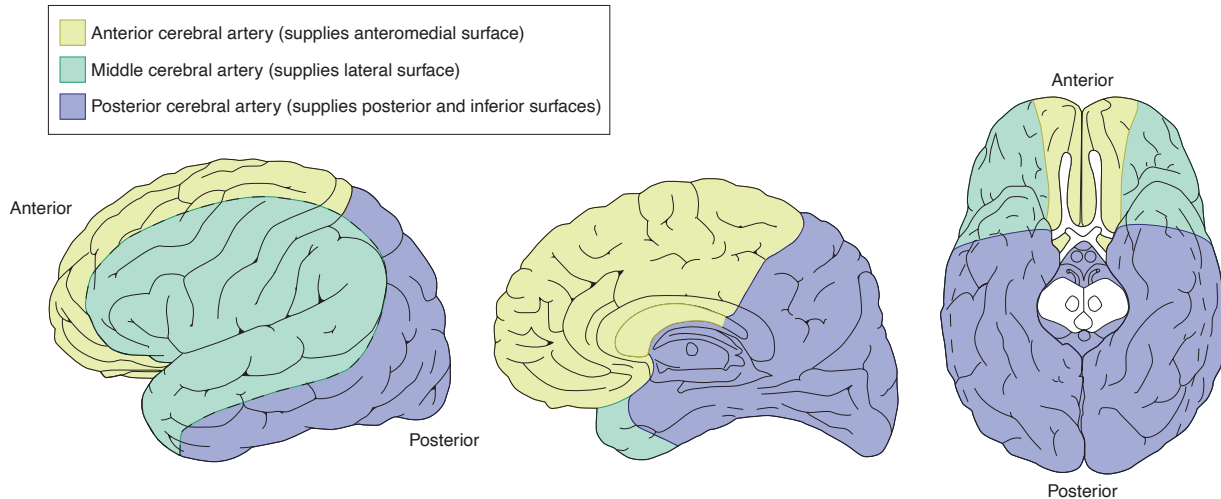
Hypoxemia increases CPP only if $P_{O_2} < 50$ mm Hg.

CPP is directly proportional to P_{CO_2} until $P_{CO_2} > 90$ mm Hg.



Homunculus

Topographic representation of motor and sensory areas in the cerebral cortex. Distorted appearance is due to certain body regions being more richly innervated and thus having ↑ cortical representation.

**Cerebral arteries—cortical distribution****Watershed zones**

Cortical border zones occur between anterior and middle cerebral arteries and posterior and middle cerebral arteries (blue areas in **A**). Internal border zones occur between the superficial and deep vascular territories of the middle cerebral artery (red areas in **A**).

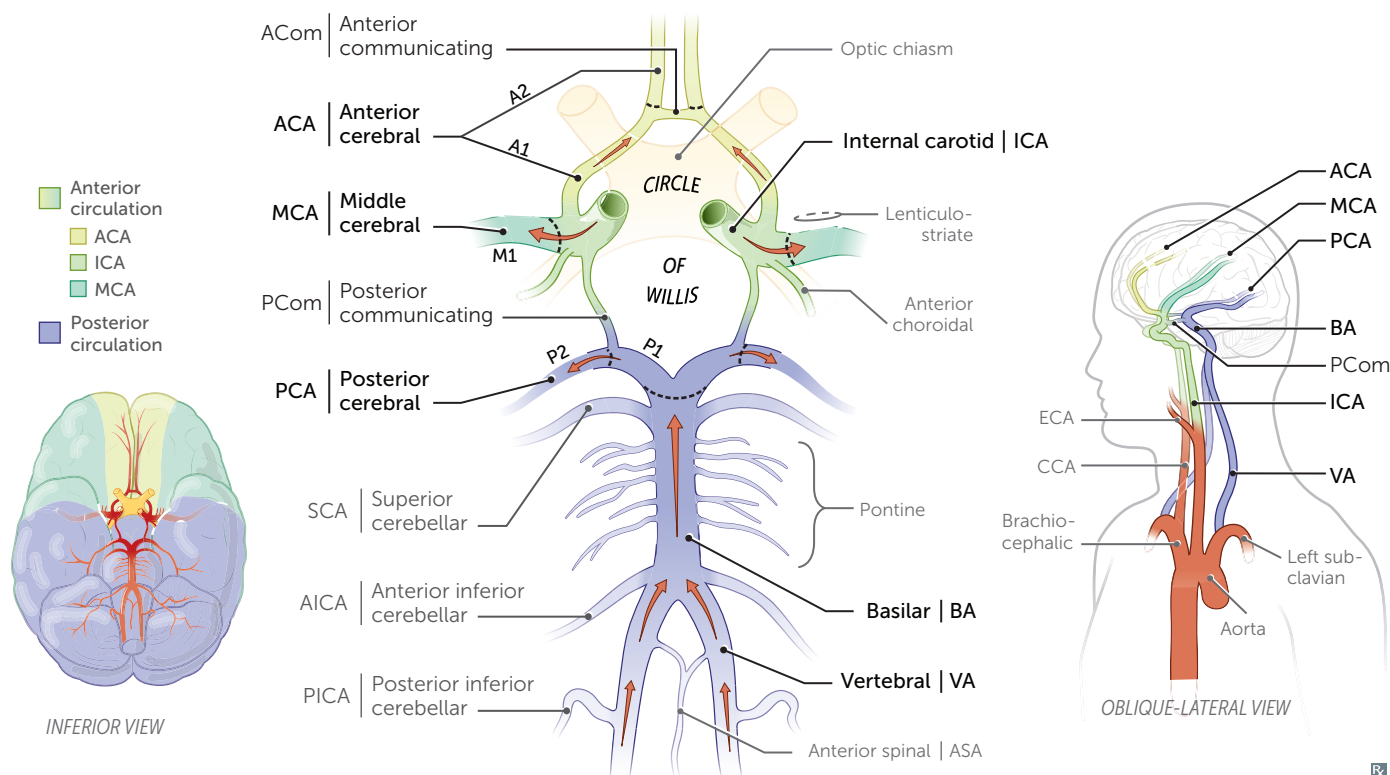
Common locations for brain metastases.

Infarct due to severe hypoperfusion:

- ACA-MCA watershed infarct—proximal upper extremity weakness sparing the lower extremities (“man-in-a-barrel syndrome”).
- PCA-MCA watershed infarct—higher-order visual dysfunction.

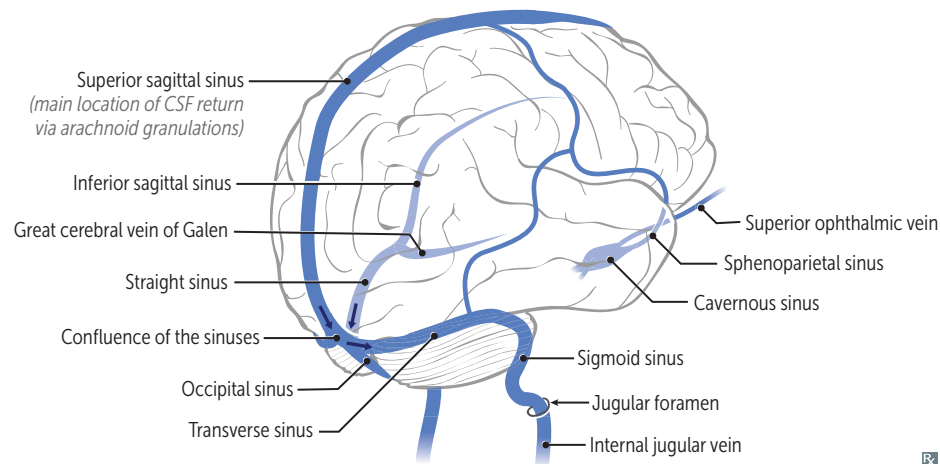
Circle of Willis

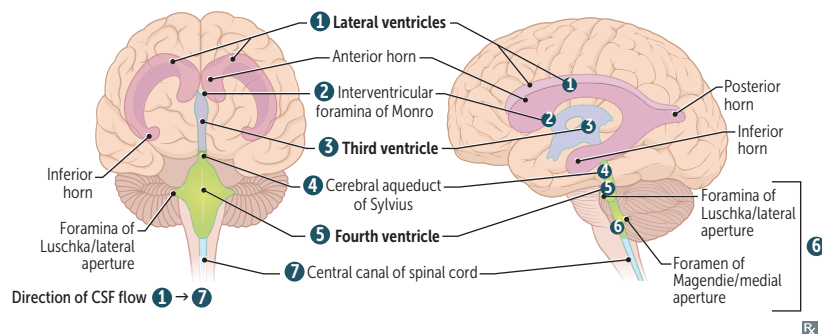
Anastomoses linking anterior and posterior circulations. Maintain cerebral perfusion in cases of occlusion/stenosis of major cranial arteries.

**Dural venous sinuses**

Large venous channels **A** that run through the periosteal and meningeal layers of the dura mater. Drain blood from cerebral veins (arrow) and receive CSF from arachnoid granulations. Empty into internal jugular vein.

Venous sinus thrombosis—presents with signs/symptoms of ↑ ICP (eg, headache, seizures, papilledema, focal neurologic deficits). May lead to venous hemorrhage. Associated with hypercoagulable states (eg, pregnancy, OCP use, factor V Leiden).



Ventricular system

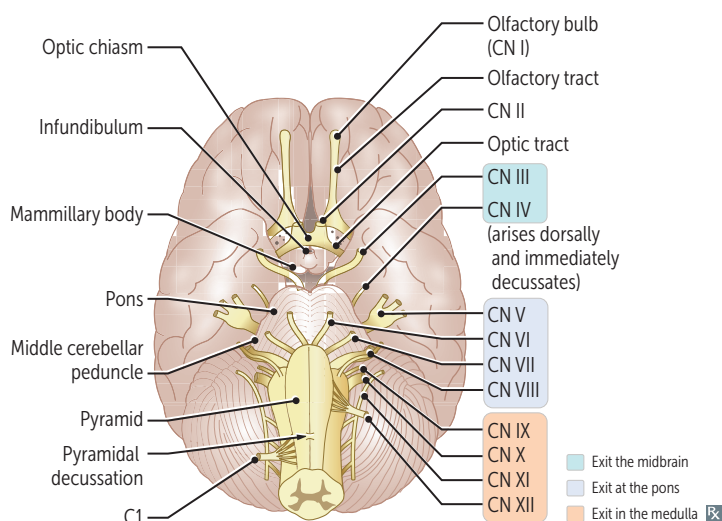
Lateral ventricles → 3rd ventricle via right and left interventricular foramina of Monro.

3rd ventricle → 4th ventricle via cerebral aqueduct of Sylvius.

4th ventricle → subarachnoid space via:

- Foramina of **L**uschka = **l**ateral.
- Foramen of **M**agendie = **m**edial.

CSF made by choroid plexuses located in the lateral, 3rd, and 4th ventricles. Travels to subarachnoid space via foramina of Luschka and Magendie, is reabsorbed by arachnoid granulations, and then drains into dural venous sinuses.

Brainstem—ventral view

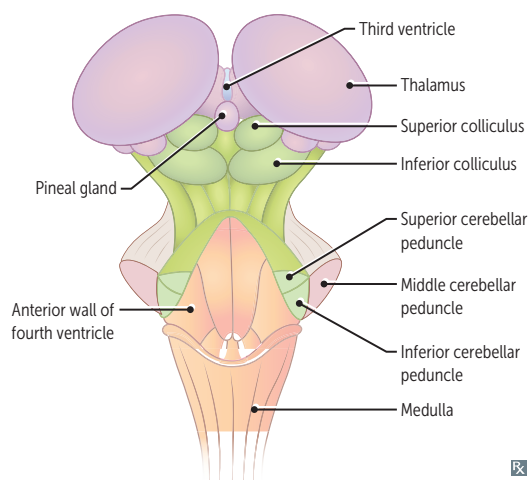
4 CN exit above pons (I, II, III, IV).

4 CN exit the pons (V, VI, VII, VIII).

4 CN exit in medulla (IX, X, XI, XII).

4 CN nuclei are medial (III, IV, VI, XII).

“Factors of 12, except 1 and 2.”

Brainstem—dorsal view (cerebellum removed)

Pineal gland—melatonin secretion, circadian rhythm.

Superior colliculi—process visual stimuli; direct eye and head movements primarily to visual stimuli.

Inferior colliculi—auditory processing; direct eye movements to auditory stimuli.

Eyes **above** ears → superior colliculus (visual)
above inferior colliculus (auditory).

Cranial nerve nuclei

Located in tegmentum portion of brainstem
(between dorsal and ventral portions):

- Midbrain—nuclei of CN III, IV
- Pons—nuclei of CN V, VI, VII, VIII
- Medulla—nuclei of CN IX, X, XII
- Spinal cord—nucleus of CN XI

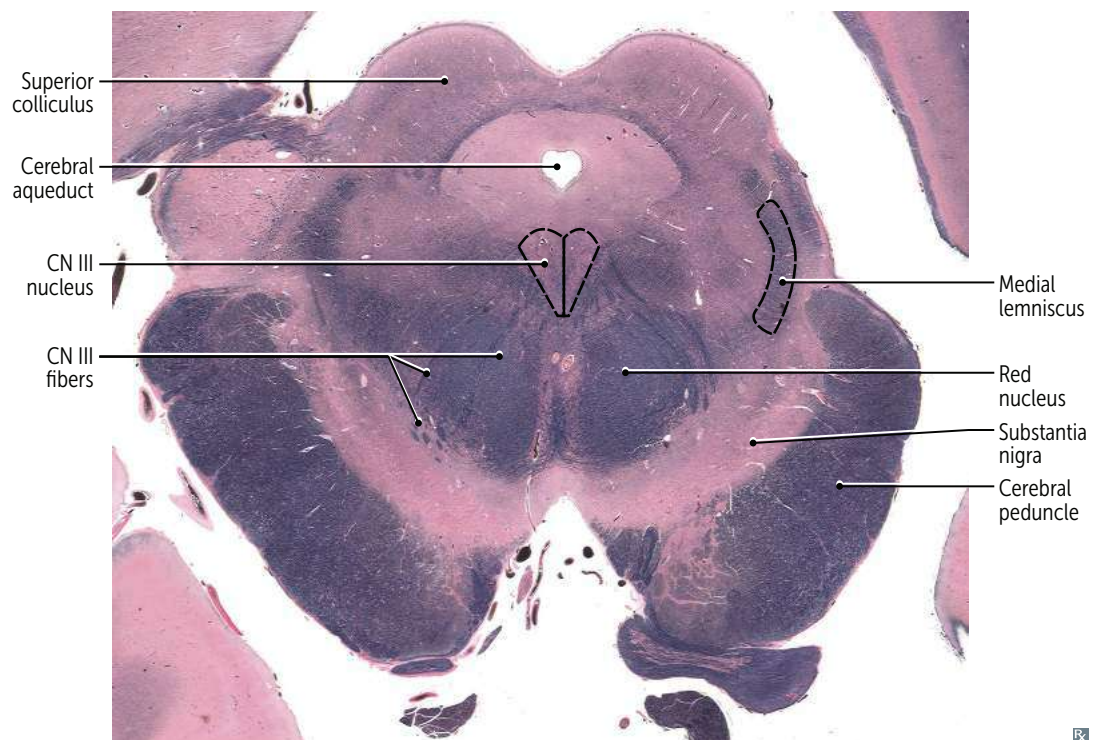
Lateral nuclei = sensory (**a**lar plate).

—Sulcus limitans—

Medial nuclei = **m**otor (basal plate).

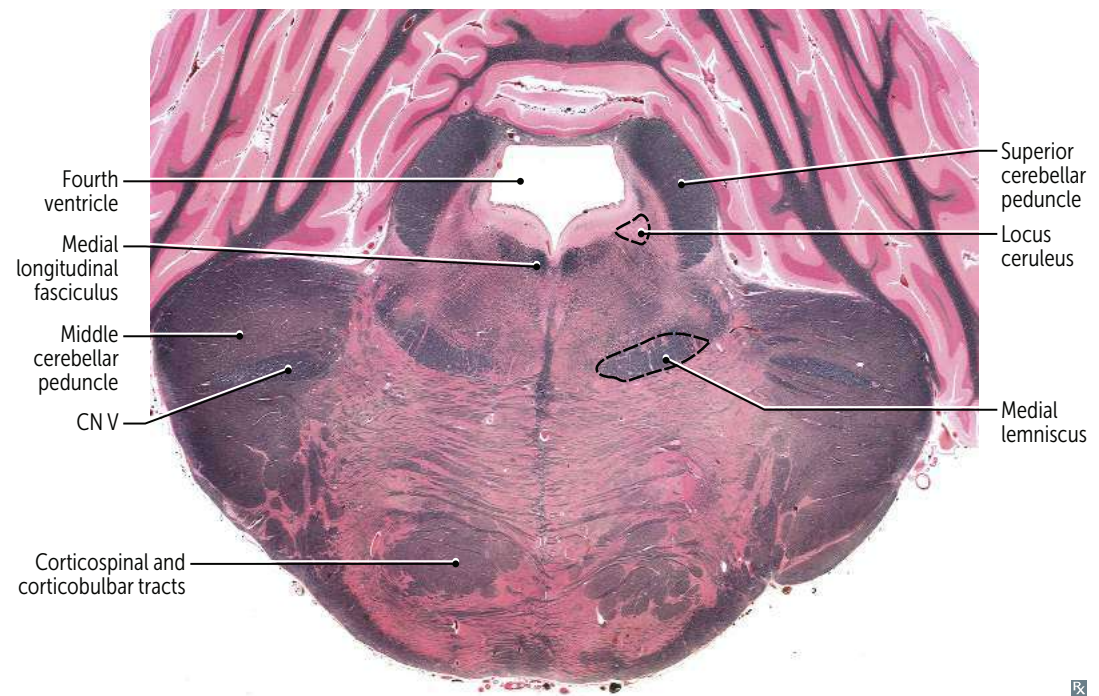
Vagal nuclei

NUCLEUS	FUNCTION	CRANIAL NERVES
Nucleus tractus solitarius	Visceral s ensory information (eg, taste, baroreceptors, gut distention) May play a role in vomiting	VII, IX, X
Nucleus ambiguus	M otor innervation of pharynx, larynx, upper esophagus (eg, swallowing, palate elevation)	IX, X
Dorsal motor nucleus	Sends autonomic (parasympathetic) fibers to heart, lungs, upper GI	X

Brainstem cross sections**Midbrain**

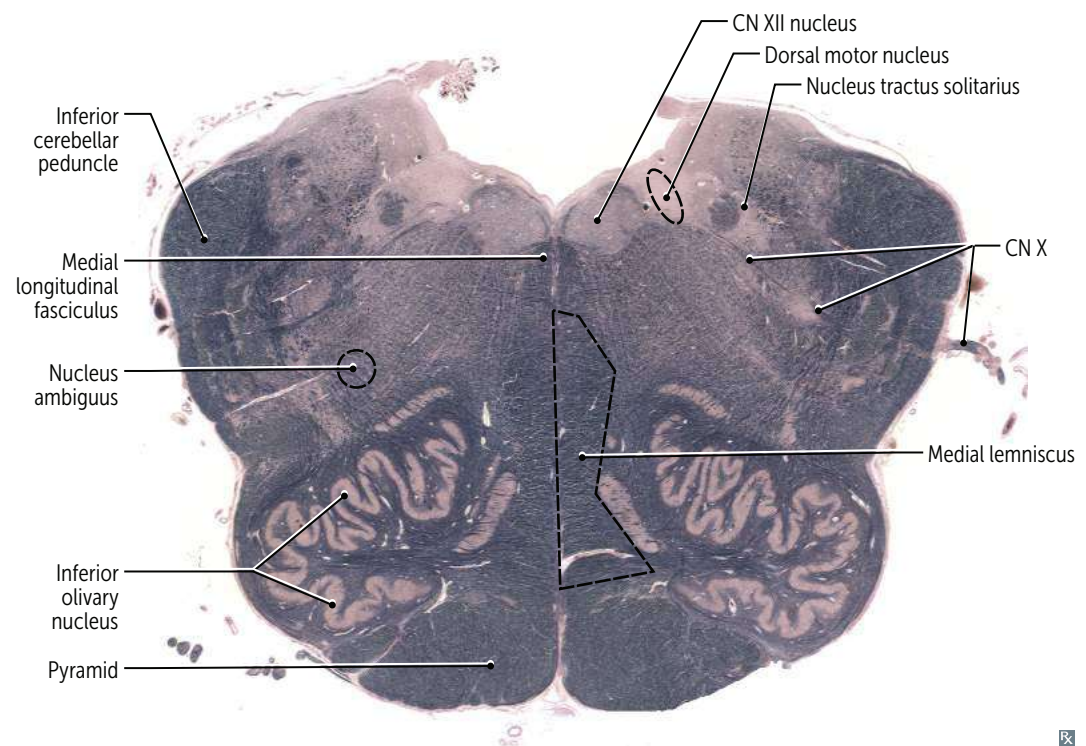
Brainstem cross sections (*continued*)

Pons



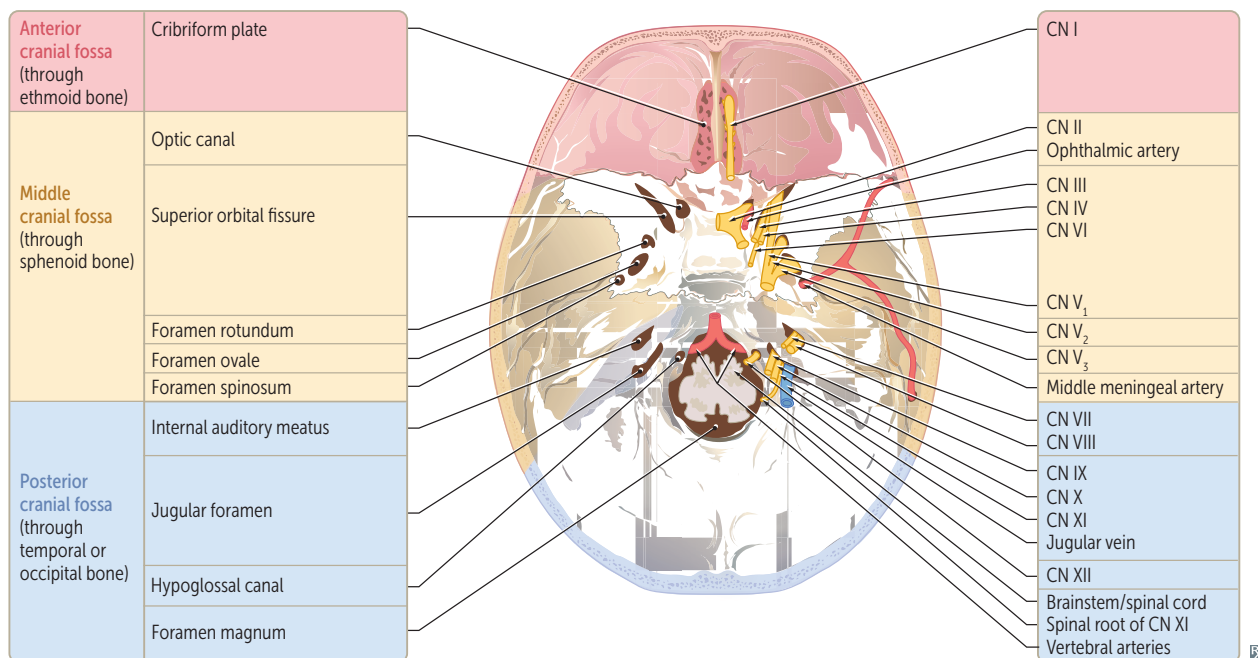
Rx

Medulla

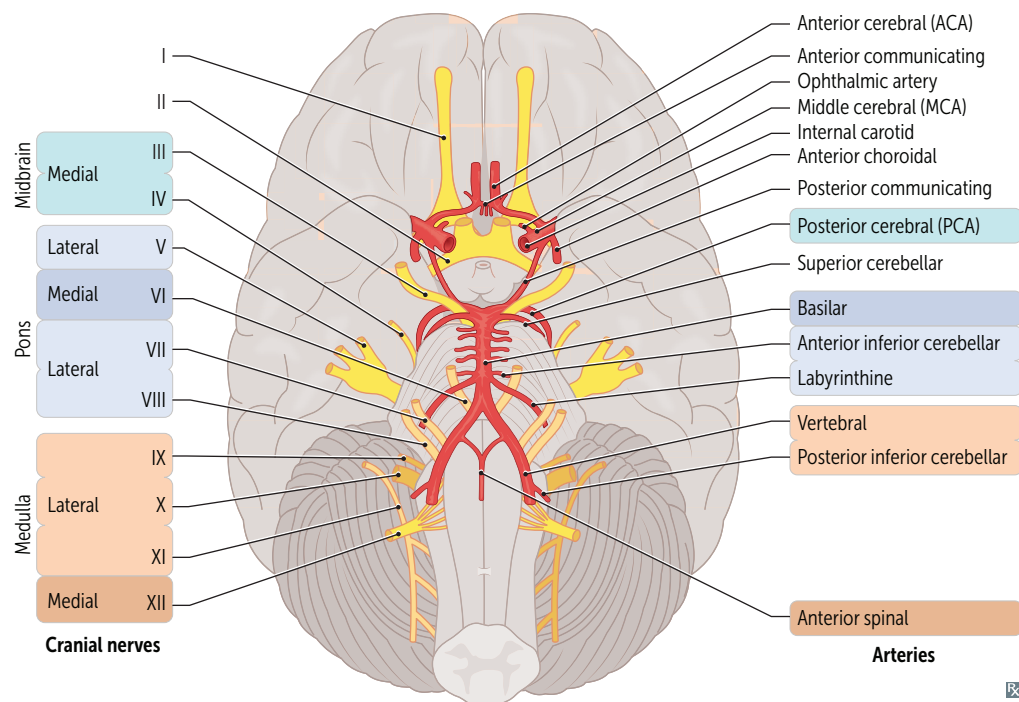


Rx

Cranial nerves and vessel pathways



Cranial nerves and arteries



Cranial nerves

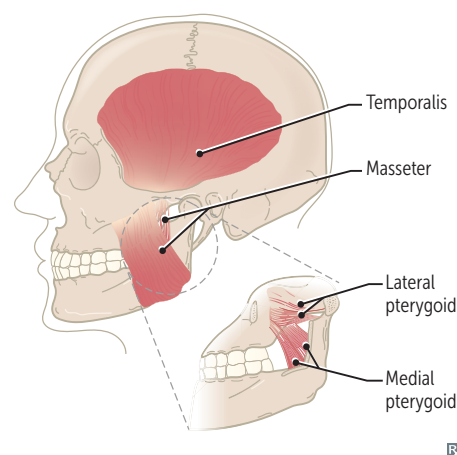
NERVE	CN	FUNCTION	TYPE	MNEMONIC
Olfactory	I	Smell (only CN without thalamic relay to cortex)	Sensory	Some
Optic	II	Sight	Sensory	Say
Oculomotor	III	Eye movement (SR, IR, MR, IO), pupillary constriction (sphincter pupillae), accommodation (ciliary muscle), eyelid opening (levator palpebrae)	Motor	Marry
Trochlear	IV	Eye movement (SO). Crosses midline → only CN with contralateral function	Motor	Money
Trigeminal	V	Mastication, facial sensation (ophthalmic, maxillary, mandibular divisions), somatosensation from anterior 2/3 of tongue, dampening of loud noises (tensor tympani)	Both	But
Abducens	VI	Eye movement (LR)	Motor	My
Facial	VII	Facial movement, eye closing (orbicularis oculi), auditory volume modulation (stapedius), taste from anterior 2/3 of tongue (chorda tympani), lacrimation, salivation (submandibular and sublingual glands are innervated by CN seven)	Both	Brother
Vestibulocochlear	VIII	Hearing, balance	Sensory	Says
Glossopharyngeal	IX	Taste and sensation from posterior 1/3 of tongue, swallowing, salivation (parotid gland), monitoring carotid body and sinus chemo- and baroreceptors, and elevation of pharynx/larynx (stylopharyngeus)	Both	Big
Vagus	X	Taste from supraglottic region, swallowing, soft palate elevation, midline uvula, talking, cough reflex, parasympathetics to thoracoabdominal viscera, monitoring aortic arch chemo- and baroreceptors	Both	Brains
Accessory	XI	Head turning, shoulder shrugging (SCM, trapezius)	Motor	Matter
Hypoglossal	XII	Tongue movement	Motor	Most

Cranial nerve reflexes

REFLEX	AFFERENT	EFFERENT
Accommodation	II	III
Corneal	V ₁ ophthalmic (nasociliary branch)	Bilateral VII (temporal and zygomatic branches—orbicularis oculi)
Cough	X	X (also phrenic and spinal nerves)
Gag	IX	X
Jaw jerk	V ₃ (sensory—muscle spindle from masseter)	V ₃ (motor—masseter)
Lacrimation	V ₁ (loss of reflex does not preclude emotional tears)	VII
Pupillary	II	III

Mastication muscles

3 muscles close jaw: **m**asseter, **t**emporalis, **m**edial pterygoid (**my** **t**ooth **m**unch).
Lateral **p**terygoids **l**ower (open) and **p**rotrude jaw.
 All are innervated by mandibular branch of trigeminal nerve (CN V₃).

**Spinal nerves**

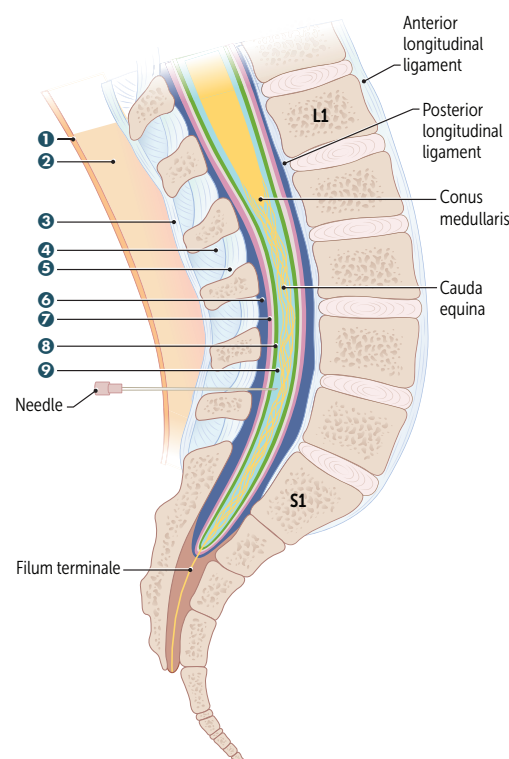
There are 31 pairs of spinal nerves: 8 cervical, 12 thoracic, 5 lumbar, 5 sacral, 1 coccygeal.
 Nerves C1–C7 exit above the corresponding vertebrae (eg, C3 exits above the 3rd cervical vertebra).
 C8 spinal nerve exits below C7 and above T1. All other nerves exit below (eg, L2 exits below the 2nd lumbar vertebra).

Spinal cord—lower extent

In adults, spinal cord ends at lower border of L1–L2 vertebrae. **S**ubarachnoid **s**pace (which contains the CSF) extends to lower border of **S2** vertebra. Lumbar puncture (LP) is usually performed between L3–L4 or L4–L5 (level of cauda equina) to obtain sample of CSF while avoiding spinal cord. To **keep** the cord **alive**, keep the spinal needle between **L3** and **L5**.

Needle passes through:

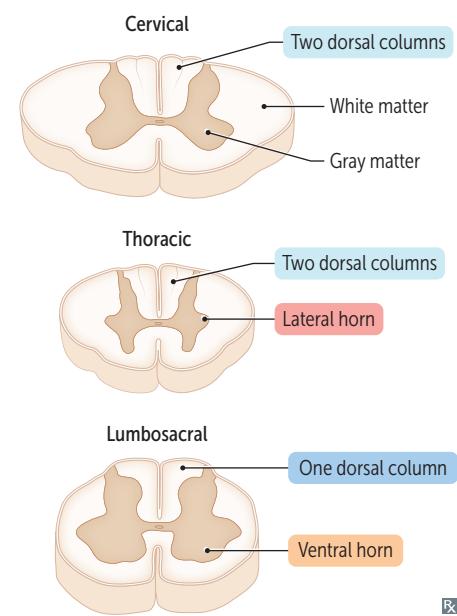
- ① Skin
- ② Fascia and fat
- ③ Supraspinous ligament
- ④ Interspinous ligament
- ⑤ Ligamentum flavum
- ⑥ Epidural space (epidural anesthesia needle stops here)
- ⑦ Dura mater
- ⑧ Arachnoid mater
- ⑨ Subarachnoid space (CSF collection occurs here)

**Conus medullaris and cauda equina syndrome**

Rare neurosurgical emergencies caused by compression (eg, disc herniation, tumors, trauma) of terminal end of spinal cord (conus medullaris) or lumbosacral spinal nerve roots (cauda equina). Present with radicular low back pain, saddle/perianal anesthesia, bladder and bowel dysfunction, lower limb weakness → symmetric with upper motor neuron (UMN) signs (eg, spastic) in conus medullaris syndrome, asymmetric with LMN signs (eg, flaccid) in cauda equina syndrome.

Spinal cord levels and associated tracts

Legs (**l**umbosacral) are **l**ateral in **l**ateral corticospinal, spinothalamic tracts. Dorsal columns are organized as you are, with hands at sides. “Arms outside, legs inside.”

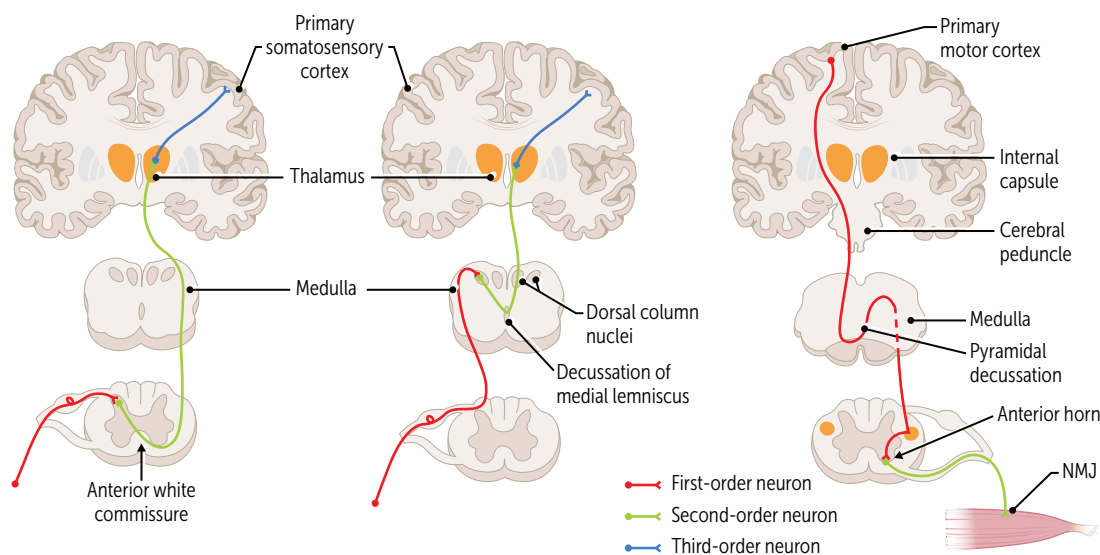


Spinal tract anatomy and functions

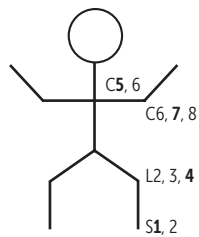
Spinothalamic tract and dorsal column (ascending tracts) **synapse** then **cross**.

Corticospinal tract (descending tract) **crosses** then **synapses**.

	Spinothalamic tract	Dorsal column	Corticospinal tract
FUNCTION	Pain, temperature	Pressure, vibration, fine touch, proprioception (conscious)	Voluntary movement
1ST-ORDER NEURON	Sensory nerve ending (A δ and C fibers) of pseudounipolar neuron in dorsal root ganglion → enters spinal cord	Sensory nerve ending of pseudounipolar neuron in dorsal root ganglion → enters spinal cord → ascends ipsilaterally in dorsal columns	UMN: 1° motor cortex → descends ipsilaterally (through posterior limb of internal capsule and cerebral peduncle), decussates at caudal medulla (pyramidal decussation) → descends contralaterally
1ST SYNAPSE	Posterior horn (spinal cord)	Nucleus g racilis (medial, lower limbs, below T6) and nucleus c uneatus (lateral, upper limbs, above T6) in the ipsilateral medulla (g ross on the ground, c louds in the sky)	Anterior horn (spinal cord)
2ND-ORDER NEURON	Decussates in spinal cord as the anterior white commissure → ascends contralaterally	Decussates in medulla → ascends contralaterally as the medial lemniscus	LMN: leaves spinal cord
2ND SYNAPSE	VPL (thalamus)	VPL (thalamus)	NMJ (skeletal muscle)
3RD-ORDER NEURON	Projects to 1° somatosensory cortex	Projects to 1° somatosensory cortex	



Clinical reflexes



Reflexes count up in order (main nerve root in bold):
Achilles reflex = S1, S2 (“buckle my shoe”)
Patellar reflex = L2-L4 (“kick the door”)
Biceps and brachioradialis reflexes = C5, C6 (“pick up sticks”)
Triceps reflex = C6, C7, C8 (“lay them straight”)

Additional reflexes:
Cremasteric reflex = L1, L2 (“testicles move”)
Anal wink reflex = S3, S4 (“winks galore”)
Reflex grading:
0: absent
1+: hypoactive
2+: normal
3+: hyperactive
4+: clonus

Primitive reflexes

Primitive CNS reflexes present in healthy infants which disappear within 1st year of life due to inhibition from developing frontal lobe. Absent in neurologically intact adult. May reemerge in adults with frontal lobe lesions → loss of inhibition.

Moro reflex	“Hang on for life” reflex—abduct/extend arms when startled, and then draw together.
Rooting reflex	Movement of head toward one side if cheek or mouth is stroked (nipple seeking).
Sucking reflex	Sucking response when roof of mouth is touched.
Palmar reflex	Curling of fingers if palm is stroked.
Plantar reflex	Dorsiflexion of large toe and fanning of other toes with plantar stimulation. Babinski sign —presence of this reflex in an adult, which may signify a UMN lesion.
Galant reflex	Stroking along one side of the spine while newborn is in ventral suspension (face down) causes lateral flexion of lower body toward stimulated side.

Landmark dermatomes

DERMATOME	CHARACTERISTICS
C2	Posterior half of skull
C3	High turtleneck shirt Diaphragm and gallbladder pain referred to the right shoulder via phrenic nerve C3, 4, 5 keeps the diaphragm alive
C4	Low-collar shirt
C6	Includes thumbs Thumbs up sign on left hand looks like a 6
T4	At the nipple T4 at the teat pore
T7	At the xiphoid process 7 letters in xiphoid
T10	At the umbilicus (belly button) Point of referred pain in early appendicitis
L1	At the Inguinal Ligament
L4	Includes the kneecaps Down on ALL 4 ’s
S2, S3, S4	Sensation of penile and anal zones S2, 3, 4 keep the penis off the floor

